Cardiovascular Drug Targets Invisible to eQTLGen Whole-Blood Expression Instruments: A Target-Level Map of Where Candidate Instruments Exist

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Abstract

Background: Human genetics validated the major lipid-lowering targets through loss-of-function carriers and biomarker-weighted variants. Druggable-genome screens rely instead on eQTLGen whole-blood cis-expression quantitative trait loci (eQTLs), known to miss a third or more of druggable genes. Which cardiovascular targets they miss, and whether other resources offer instruments, is unmapped.

Methods: We classified 726 cardiovascular drug targets by whether eQTLGen whole blood supplies a cis instrument ($P<5\times10^{-8}$, gene body ±100 kb). For each target without one, a pre-written rule asked whether GTEx v8 tissues or 3 published plasma cis-protein QTL tables hold a qualifying variant (a candidate instrument).

Results: No eQTLGen whole-blood instrument existed for 280 targets (38.6%), with no evidence of a phase gradient. They included 6 of 10 lipid targets with landmark human genetic evidence (*PCSK9*, *APOC3*, *ANGPTL3*, *APOB*, *NPC1L1*, *LPA*; *HMGCR*, *CETP*, *LDLR* and *LPL* were instrumented) and 12 of 33 targets of approved cardiovascular drugs. Of the 280, 151 (53.9%) had a candidate in a cardiovascular-relevant tissue or plasma protein, 72 (25.7%) only in another tissue, and 57 (20.4%) none. Heart failure and cardiomyopathy targets were most often without one (21 of 70); excluding microRNA and mitochondrial genes, 11 of the 16 protein-coding genes had never been measured by a plasma proteomic study. Of 5 candidates tested as positive controls, 2 reproduced the drug's direction.

Conclusions: eQTLGen whole-blood expression screens are blind to most of the lipid targets that genetics validated. Where no circulating biomarker offers another route, as in heart failure, the gap reflects measurement coverage.

Key Words: Mendelian randomization analysis; quantitative trait loci; drug discovery; cardiovascular agents; proteomics

Nonstandard Abbreviations and Acronyms

eQTL, expression quantitative trait locus; GTEx, Genotype-Tissue Expression project; MR, Mendelian randomization; pQTL, protein quantitative trait locus; TPM, transcripts per million; UKB-PPP, UK Biobank Pharma Proteomics Project.

Introduction

The case for human genetics in cardiovascular drug discovery was made at a handful of lipid genes. Carriers of loss-of-function variants in *PCSK9* have low low-density lipoprotein cholesterol and less coronary disease;[1] the same holds for inactivating variants in *NPC1L1*,[2] *APOC3*[3,4] and *ANGPTL3*,[5] and common variants that raise lipoprotein(a) raise coronary risk.[6] Genetic scores built from variants in *PCSK9* and *HMGCR*, scaled to their effect on cholesterol, were used to anticipate the effects of inhibiting each target.[7] In each case the instrument was a coding variant or a variant selected for its effect on a circulating biomarker. More broadly, drug mechanisms with genetic support succeed in development more often than those without.[8–12]

Systematic screens work differently. To cover thousands of genes at once, drug-target Mendelian randomization (MR) takes variants that alter the expression or protein level of the target as instruments,[13–15] and the most widely used source is eQTLGen, a whole-blood cis-eQTL meta-analysis of 31,684 people that tested 19,250 autosomal genes expressed in blood.[16] That many drug targets fall outside that set is in print, although as a methods step and not as a finding. Storm and colleagues found blood eQTLs for 2,786 of 4,863 druggable genes.[17] Of the 4,479 druggable genes defined by Finan and colleagues,[18] 2 later studies retrieved eQTLGen data for 2,888 and 2,525,[19,20] and a third, adding lung, for 2,531 of 4,317.[21] Sadler and colleagues counted about 12,550 of 19,150 autosomal protein-coding genes as testable by blood eQTL and showed *HCN4* as an approved-drug target with no eQTL.[22] With 49 GTEx tissues, Gaziano and colleagues covered 1,016 of 1,263 targets of approved or clinical-stage drugs.[23] The absence of the X chromosome from eQTLGen has been noted.[24,25] We do not present the overall share as new.

What is missing is the target-level picture for cardiovascular medicine. No study reports which cardiovascular targets are affected, by clinical phase or by name, or asks for each one whether another resource offers even a candidate instrument. The answer matters for how negative results are read: a target with no instrument is silently absent from an expression-based screen and is easily counted as lacking genetic support.[26,27] It matters most where no circulating biomarker offers another route to an instrument.

We set out to (1) identify the cardiovascular drug targets that an eQTLGen whole-blood expression screen cannot test, by clinical phase and by name; (2) map, for each, where a candidate instrument exists in other tissues or for the plasma protein; and (3) test a small set of those candidates against known drug effects.

Methods

Data and Code Availability

All inputs are public summary-level data. The analysis code, rule documents, per-target classification, recovery map and number trace are available at https://github.com/Rugved3-droid/causal2drug. Because the study used only published, de-identified summary statistics and involved no human participants, institutional review board approval and informed consent were not required. Reporting follows STROBE-MR.[28]

Target Set

Targets, indications and clinical phases came from the target–indication table of Minikel et al.[10] After removing duplicate rows and 4 programmes with conflicting phase labels, 29,456 target–indication pairs remained (2,517 targets, 1,014 indications). The primary cardiovascular set comprised every indication that the source tags as cardiovascular, plus indications matching 8 lipid terms: 93 indications, 2,128 pairs and 726 targets (Table 1, Figure 1). No indication was excluded after the rule was set. Three alternative definitions (630, 663 and 594 targets) are sensitivity analyses. A target's phase is the highest phase reached across its indications in the set; phase labels were used as published. Targets were grouped into 10 indication areas, and a target was counted in every area in which it has a programme.

Instrument Status in eQTLGen

Gene coordinates came from Ensembl GRCh37 release 87; symbols without a unique record were resolved through their HGNC record.[29] Targets were joined to eQTLGen phase I by Ensembl gene identifier. An instrument was defined as at least 1 cis-eQTL with $P<5\times10^{-8}$ within the gene body ±100 kb. Each target was assigned 1 status. *Instrumented*: at least 1 such variant. *No eQTL data*: the gene is not among the 19,250 genes eQTLGen tested (autosomal gene not tested; X-linked gene; or mitochondrial gene, eQTLGen having analysed neither the X chromosome nor the mitochondrial genome). *No qualifying instrument*: tested, but no variant meets the definition. Targets in the last 2 statuses are called uninstrumentable; the first of the 2 is a lower bound and their sum an upper bound. Windows of ±500 kb and ±1 Mb were sensitivity analyses.

Approved-Drug Verification

Because a phase label attaches to a target–indication pair and not to a drug, launched and phase III targets were checked against ChEMBL release 37.[30] A target was counted as having an approved cardiovascular drug if a drug linked directly to the gene (single protein, specific protein complex, or the messenger RNA for antisense and RNA-interference drugs) has an approved indication that sits only in cardiovascular disease branches of MeSH, and has an approval record at the Food and Drug Administration (FDA) or the European Medicines Agency (EMA). *Approved* therefore means ever approved, whether or not the drug is still marketed. A permissive rule (any ChEMBL link, any cardiovascular indication)

was applied alongside it. The 327 ChEMBL evidence rows for the 32 targets on which the 2 rules disagreed were reviewed by the authors; every disagreement arose from a drug recorded against a receptor family or complex group, or from an indication outside cardiovascular branches, and the strict rule was retained. The site of drug action was taken from the mechanism section of the label (Table S23).

Recovery Map of Candidate Instruments

For every uninstrumentable target we asked whether a variant meeting the same definition exists elsewhere. We call such a variant a *candidate instrument*: it satisfies a statistical definition, and nothing more was assumed about it. The rule was written before any GTEx or pQTL result was opened; it is a dated protocol document without an external timestamp (Supplemental Methods). The definition was unchanged: $P<5\times10^{-8}$ within the gene body ±100 kb, in each resource's own genome build and on top of that resource's own significance flag.

Tissue expression. GTEx v8 significant variant–gene pairs for all 49 tissues were searched by Ensembl gene identifier.[31] Eight tissues were designated cardiovascular-relevant in advance: heart left ventricle, heart atrial appendage, aorta, coronary artery, tibial artery, liver, kidney cortex and adrenal gland. GTEx whole blood did not count.

Plasma protein. The published pQTL tables of the UK Biobank Pharma Proteomics Project (UKB-PPP; 54,219 participants), deCODE (35,559) and Fenland (10,708) were searched.[32–34] A target had a protein candidate if a study labelled a pQTL for its protein as cis, with $P<5\times10^{-8}$, and the variant lay within the gene body ±100 kb. Only published tables were used, not full summary statistics.

Targets were then placed in tiers: a candidate in a cardiovascular-relevant tissue or a plasma cis-pQTL (*CV-relevant or plasma-protein candidate*); a candidate only in other tissues (*other-tissue candidate only*); and *locked*, meaning none in the resources checked, divided into *never assayed* (measured by none of the 3 proteomic studies) and *assayed, no instrument*. The tiers, and an exception for 3 genes whose Ensembl identifier changed between builds, were set after the first results were seen and are post hoc. A second implementation, written in a separate AI-assisted session from the rule documents alone, without access to the original code or per-target outputs, reproduced all target classifications and tiers; the discrepancies it found, a P-value underflow that changed some recorded lead variants and 2 assay-coverage entries, were corrected and changed no classification (Supplemental Methods).

Statistics and Controls

Expression was taken from GTEx (version 10 median transcripts per million, TPM) and tissue specificity from the Human Protein Atlas.[31,35] Proportions are given with Wilson 95% CIs and the difference between launched and preclinical targets with a Newcombe 95% CI; all tests are descriptive. Five candidates were tested as positive controls under a protocol written before any outcome association was retrieved: *PCSK9* on low-density lipoprotein

cholesterol, *ANGPTL3* and *APOC3* on triglycerides, and *AGTR1* and *REN* on systolic blood pressure. The protocol fixed the expected sign from the drug's effect, the instrument (the lead qualifying variant in the recovery map, a plasma pQTL where one existed), the outcome study (largest European-ancestry GWAS Catalog study with harmonised, indexed statistics), the harmonization code and the estimator (single-variant Wald ratio). A control passed if F≥10, MR P<0.05 and the sign was as specified; a variant that could not be harmonized was reported, not replaced. Three earlier controls with eQTLGen instruments are reported in the Supplemental Methods.

Results

eQTLGen Whole-Blood Instruments Miss Most of the Best-Validated Lipid Targets

Of 726 cardiovascular targets, 443 (61.0%) were instrumented in eQTLGen whole blood (Figure 1). No eQTL data existed for 217 (29.9%) and 63 (8.7%) were tested without a qualifying instrument, so 280 (38.6%) were uninstrumentable; 3 had no resolvable gene record. This is the size of gap that druggable-genome studies have reported,[17,19,20] and it was stable across set definitions and cis-windows (Tables S3 and S4).

Of 10 lipid targets with landmark human genetic evidence, a list assembled from the literature after the analysis,[1–7] 6 were uninstrumentable. *PCSK9*, *APOC3*, *ANGPTL3*, *APOB* and *LPA* were not among the genes eQTLGen tested, and *NPC1L1* was tested without a qualifying variant. *HMGCR*, *CETP*, *LDLR* and *LPL* were instrumented. GTEx whole blood, a smaller resource, held a qualifying variant for *PCSK9* (Table 2). An eQTLGen whole-blood expression screen would therefore have been silent on most of the targets that established the field.

The share with no eQTL data was 30.0% of preclinical targets (95% CI, 25.3–35.2), 28.6% of phase I (20.3–38.6), 33.3% of phase II (26.7–40.7), 18.7% of phase III (11.5–28.9) and 34.8% of launched targets (24.5–46.9; Figure 2, Table S11). Launched and preclinical targets differed by 4.8 percentage points (95% CI, −6.8 to 17.8). We found no evidence of a phase gradient, although the intervals do not exclude a difference of several points in either direction.

Approved-Drug Targets Affected

Thirty-three launched targets had an approved cardiovascular drug as defined. Nine had no eQTL data (27.3%) and 3 had no qualifying instrument, 12 in all (36.4%). Eleven are named as examples: *AGTR1*, *ANGPTL3*, *APOB*, *APOC3*, *AVPR2*, *F2*, *PCSK9*, *REN* and *SLC5A1* had no eQTL data, and *ITGA2B* and *NPC1L1* were tested without a qualifying instrument (Table 2). The twelfth, *IL1B*, is in Table 2 but not named: its drug binds both interleukin-1α and interleukin-1β and is approved for pericarditis, whereas the source labels the target launched for atherosclerosis. Ten of the 11 have a currently marketed drug; the exception is *APOB*, whose only linked drug, mipomersen, is discontinued (Table S18).

Missing Targets Are Not Expressed in Blood

eQTLGen tested genes expressed in blood, so targets without data should be those that are not. That is what we found: of the 188 autosomal targets with no eQTL data, 182 had a GTEx expression value and 180 had whole-blood expression below 1 TPM, and tissue-restricted expression was far more common than among instrumented targets (odds ratio, 7.47 [95% CI, 4.12–13.55]; Table S9). The other 29 were on chromosomes eQTLGen did not analyse: 27 X-linked and 2 mitochondrial. Twelve of the 27 X-linked targets were expressed in blood at 1 TPM or more (Table S10), so their absence reflects the resource and not biology.

Recovery Map of Candidate Instruments

Of the 280 uninstrumentable targets, 223 (79.6%) had a candidate instrument in at least 1 checked resource and 57 (20.4%) were locked (Table 3, Figure 3). GTEx held a candidate in a tissue other than blood for 199 targets, and a plasma cis-pQTL table for 95 (Table S16).

For 151 targets (53.9%) the candidate was in a cardiovascular-relevant tissue or a plasma protein table: 56 through tissue alone, 60 through protein alone and 35 through both. Tibial artery (44 targets), aorta (34) and atrial appendage (29) contributed most; kidney cortex, the smallest tissue sample, contributed none.

For 72 targets (25.7%) the only candidate lay in a tissue outside the pre-specified list, most often testis (21 targets), tibial nerve (17), cultured fibroblasts (16) and subcutaneous adipose (14). Twenty-seven of the 72 rested on a single tissue and 7 on testis alone (Table S14).

Fifty-seven targets were locked. Nine are microRNA genes and 2 mitochondrial genes, which no plasma protein panel could measure. Of the other 46, all protein-coding genes, 24 had never been assayed by any of the 3 proteomic studies and 22 had been assayed without a qualifying cis-pQTL (Table S15). Most X-linked targets had a candidate in resources that analyse the X chromosome (23 of 27).

Heart Failure and Cardiomyopathy Stand Apart

Locked targets made up 27.0% of preclinical and 3.7% of launched uninstrumentable targets (Figure 3A). By indication area the contrast was sharper (Figure 3B). Of 70 uninstrumentable heart failure and cardiomyopathy targets, 25 (35.7%) had a CV-relevant or plasma-protein candidate, 24 (34.3%) an other-tissue candidate only, and 21 (30.0%) were locked. Four are microRNA genes and 1 is a mitochondrial gene; of the other 16, all protein-coding genes, 11 had never been measured by a plasma proteomic study. The locked set includes *DMD* and *PLN*. In thrombosis, by contrast, none of 22 targets was locked. Areas overlap because a target is counted wherever it has a programme.

The Named Approved-Drug Targets

Of the 11 named targets (Table 2), 7 had a CV-relevant or plasma-protein candidate. For *ANGPTL3*, *APOC3*, *PCSK9* and *REN* it came from plasma protein only; *AGTR1*, *APOB* and *F2* had one in a cardiovascular-relevant tissue. Three had a candidate only in other tissues: *SLC5A1* (thyroid, spleen, tibial nerve), *ITGA2B* (thyroid) and *NPC1L1* (subcutaneous adipose, testis). The X-linked vasopressin V2 receptor gene *AVPR2* was locked: it was tested in 45 GTEx tissues without a qualifying variant and measured by no proteomic study. Nine of the 11 had a GTEx tissue matching the labelled site of drug action, and only 1 (*AGTR1*, tibial artery) had a candidate in it; liver held none for *PCSK9*, *ANGPTL3*, *APOC3* or *APOB* (Table S23).

Of 4 phase III targets with an approved drug in ChEMBL, *AVPR1A*, *SLC6A2* and *PDE3A* had a CV-relevant or plasma-protein candidate. *HCN4* was locked: no candidate in any of the 25 GTEx tissues in which it was tested, including both heart tissues, and no measurement in any proteomic study.

Candidate Instruments as Positive Controls

Two controls passed, 1 failed and 2 could not be harmonized (Tables S19 and S22). The *PCSK9* plasma pQTL was rs11591147, the p.Arg46Leu loss-of-function variant;[1] it gave 0.387 SD of low-density lipoprotein cholesterol per SD higher plasma PCSK9 (95% CI, 0.377–0.396). The *APOC3* plasma pQTL was rs964184, a long-recognised triglyceride signal in the *APOA5–APOA4–APOC3–APOA1* cluster that is not specific to *APOC3*; it gave 1.171 SD of log-transformed triglycerides per SD higher plasma apolipoprotein C-III (95% CI, 1.153–1.190). The *AGTR1* tibial-artery eQTL failed: −0.037 mm Hg of systolic blood pressure per unit of normalized expression (95% CI, −0.268 to 0.194; P=0.75). The *ANGPTL3* lead pQTL is an insertion–deletion variant and the *REN* lead pQTL, of 0.8% frequency, is absent from the outcome study; neither was replaced. Exposure and outcome samples overlapped for *PCSK9* (UK Biobank contributes to both), but with F statistics of 2,305 and 115 for the passing controls the resulting bias should be small.[36] Of the 3 earlier eQTLGen controls, *HMGCR* passed, *IL6R* failed on its pre-specified sign, and *PCSK9* could not be tested because it is absent from eQTLGen (Supplemental Methods).

Discussion

eQTLGen whole-blood expression instruments, the default of druggable-genome screens, cannot see *PCSK9*, *APOC3*, *ANGPTL3*, *APOB*, *NPC1L1* or *LPA*: 6 of the 10 lipid targets with landmark human genetic evidence. They miss 12 of 33 targets of approved cardiovascular drugs, and 280 of 726 cardiovascular targets overall. The size of that gap is known for the druggable genome.[17,22,23] What this study adds is which cardiovascular targets fall in it and, for each, where a candidate instrument can and cannot be found.

The lipid targets show why the gap has not obstructed cardiovascular genetics so far, and where it will. Their genetics was never read from blood expression. It came from loss-of-function carriers and from variants weighted by their effect on a circulating biomarker,[1–3,5,7]

an approach that needs no eQTL and remains the standard for lipid and blood-pressure targets. The expression gap is therefore harmless where such a biomarker exists and decisive where it does not. Heart failure and cardiomyopathy are the clearest case. There is no circulating trait that reports the activity of most of these targets, so an instrument must come from expression in the relevant tissue or from the protein itself, and this is the area in which we most often found neither.

We read that result as a gap in measurement coverage, not as evidence that these targets are intractable. Of the 16 protein-coding genes among the locked heart failure targets (microRNA and mitochondrial genes aside), 11 have never been measured by a plasma proteomic study, and the GTEx heart samples are about 80 times smaller than eQTLGen. The locked list includes *DMD* and *PLN*, genes whose causal role in cardiomyopathy is established by Mendelian genetics.[37,38] They are the plainest demonstration that no instrument does not mean no genetic support. *HCN4*, which Sadler and colleagues showed to lack a blood eQTL,[22] is locked in every resource we checked although ChEMBL lists ivabradine as an approved drug acting on it, and *AVPR2* is in the same position. For such targets an MR study cannot return evidence for or against the mechanism, and reporting them as unsupported would misstate what was done. The never-assayed protein-coding targets (Table S15) are a short list for future proteomic panels and for expression studies in larger collections of heart, vascular and kidney tissue. X-linked targets are a different case: most had a candidate elsewhere, so their absence from eQTLGen is an analytic omission,[24,25,39] not a biological barrier.

Where a candidate sits matters. Four of every 5 uninstrumentable targets have a qualifying variant somewhere, in line with the multi-tissue coverage reported by Gaziano and colleagues,[23] but for a quarter it is only in a tissue such as testis, tibial nerve or cultured fibroblasts. eQTL effects differ in size and direction across tissues, including for lipid-lowering targets,[13] and regulatory variants found in one tissue are not interchangeable with those acting in another.[40,41] Among the named approved-drug targets, liver, the matching tissue for 4 of them, supplied no candidate, and only *AGTR1* had one in a tissue matching the drug's site of action.

An available cis-QTL is not a validated instrument. It does not establish that the variant acts on the intended target and not a neighbouring gene, that the change it marks is functionally relevant, that its direction corresponds to what the drug does, or that the association is free of confounding by linkage disequilibrium, of horizontal pleiotropy, or of assay artefacts such as altered antibody or aptamer binding.[13–15,27,42] Our controls are illustrations of this, not a validation. The two that passed rested on variants the field already knows, a *PCSK9* coding variant and a multi-gene triglyceride locus. Two lead pQTLs could not be used at all. The two failures were mismatches between what the instrument measures and what the drug does: an arterial *AGTR1* eQTL for drugs that block receptor function, and blood *IL6R* messenger RNA for a receptor that exists in membrane-bound and soluble forms,[43] for which the established instrument is the Asp358Ala variant scaled by its effect on C-reactive protein.[44] With 5 candidate controls, the pattern that plasma pQTLs passed and the tissue eQTL failed is illustrative, not general.

The findings bear on how genetic support is counted. Estimates of the value of genetic evidence rest on whether a target has support,[8–10] and integrative methods already note that missingness across data types is not random.[26] We found no evidence that the gap varies with clinical phase, but we did not test whether it biases comparisons of success between supported and unsupported targets.

Limitations

First, "locked" is relative to the resources checked and is conservative. Published pQTL tables list only associations that passed each study's own threshold ($P<1.7\times10^{-11}$ in UKB-PPP); full summary statistics, newer proteomic panels and non-plasma proteomics were not used. Second, GTEx v8 tissue samples are small and unequal: 73 for kidney cortex, 208 for liver, 213 for coronary artery, 233 for adrenal gland, 372 and 386 for the 2 heart tissues, 387 for aorta and 584 for tibial artery, against 31,684 in eQTLGen (Table S20). The tiers are confounded with tissue sample size, which we did not model. Third, all the resources are predominantly of European ancestry. Fourth, the analysis concerns eQTLGen phase I and common cis variants at 1 threshold; trans-eQTLs, splicing QTLs, rare variants and cell-type-specific or stimulated-state eQTLs were out of scope, as were biomarker-weighted instruments, which we did not enumerate per target. Fifth, the reporting tiers and the identifier-change exception were set after the first results were seen, and the protocol documents are dated but not externally timestamped; the project was not registered. "Other tissue" refers to a fixed tissue list, not to a judgement about each drug. Sixth, validation of the MR pipeline is limited: 3 of 8 controls passed, 2 failed and 3 could not be run. The *PCSK9* control exposure came from the combined, multi-ancestry UKB-PPP cohort, whereas the outcome study was European only (Table S22). No conclusion here rests on an MR estimate. Seventh, phase labels were taken as published; counts are of genes, not drugs; non-selective drugs recorded against a receptor family were not assigned to individual genes; and "approved" means ever approved. Eighth, indication areas overlap, and 46 of 93 indications fall in the "other" area.

Conclusions

An eQTLGen whole-blood expression screen is blind to most of the lipid targets that genetics validated and to a third of approved cardiovascular drug targets, with no evidence that this varies by clinical phase. For most of the affected targets a candidate instrument exists elsewhere, for a quarter only in a tissue unrelated to the indication, and for a fifth not at all. The gap matters least where a circulating biomarker offers another route and most where none does, as in heart failure, where it reflects what has not yet been measured. Candidate instruments need validation before use, and reports of drug-target MR should state which targets were untestable and keep "no instrument" distinct from "no support".

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Disclosures

None.

Supplemental Material

Supplemental Methods Tables S1–S23 Figures S1–S2 STROBE-MR checklist

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Figure Legends

Figure 1. Derivation of the cardiovascular target set and instrument status in eQTLGen whole blood. Target–indication pairs from Minikel et al. were reduced to 726 targets with a cardiovascular programme. Each was classified by whether eQTLGen provides a cis-eQTL instrument ($P<5\times10^{-8}$, gene body ±100 kb). The bracket marks the lower and upper bound of uninstrumentable targets.

Figure 2. Instrument status by highest clinical phase. Share of targets in each status. Solid orange, no eQTL data, autosomal gene not tested; hatched, X-linked or mitochondrial gene not analysed. The black tick marks the upper bound. The lower 2 rows show launched targets with an approved cardiovascular drug, by the primary definition (an approval record at the Food and Drug Administration or the European Medicines Agency) and by the permissive ChEMBL rule.

Figure 3. Recovery map of candidate instrument availability for the 280 targets with no eQTLGen whole-blood instrument. A, By highest clinical phase. **B**, By indication area, ordered by the share locked; a target is counted in each area in which it has a programme. A candidate is a variant at $P<5\times10^{-8}$ within the gene body ±100 kb; its validity as an instrument was not tested. Blue, a candidate in 1 of 8 pre-specified cardiovascular-relevant GTEx v8 tissues or a plasma cis-pQTL in UKB-PPP, deCODE or Fenland. Green, a

candidate only in other tissues. Orange, locked (no candidate in any checked resource), divided by whether any proteomic study measured the protein. Numbers in bars are targets.

Tables

Table 1. Derivation of the cardiovascular target set and its composition

	N			
Rows in the source target-indication table	29476			
Distinct target-indication pairs	29460			
Pairs after removing 4 programmes with conflicting phase labels	29456			
Distinct targets / indications in those pairs	2517 / 1014			
	Primary set	Sensitivity 1 (original 630)	Sensitivity 2	Sensitivity 3 (MeSH C14 only)
Cardiovascular indications (MeSH terms)	93	44	62	37
Cardiovascular target-indication pairs	2128	1543	1879	1364
Cardiovascular targets	726	630	663	594
Instrumented	443 (61.0%)	391 (62.1%)	409 (61.7%)	373 (62.8%)
No eQTL data (lower bound)	217 (29.9%)	180 (28.6%)	191 (28.8%)	164 (27.6%)
autosomal gene not tested	188 (25.9%)	157 (24.9%)	167 (25.2%)	144 (24.2%)
X-linked, not analysed by eQTLGen	27 (3.7%)	21 (3.3%)	22 (3.3%)	18 (3.0%)
mitochondrial,	2 (0.3%)	2 (0.3%)	2 (0.3%)	2 (0.3%)

	N			
not analysed by eQTLGen				
No qualifying instrument	63 (8.7%)	57 (9.0%)	60 (9.0%)	55 (9.3%)
Uninstrumentable, upper bound	280 (38.6%)	237 (37.6%)	251 (37.9%)	219 (36.9%)
Unresolved gene record	3 (0.4%)	2 (0.3%)	3 (0.5%)	2 (0.3%)
Autosomal-only lower bound	188 / 697 (27.0%)	157 / 607 (25.9%)	167 / 639 (26.1%)	144 / 574 (25.1%)
Highest phase reached in the set				
Preclinical	323	297	300	294
Phase I	91	90	93	85
Phase II	171	140	155	121
Phase III	75	51	60	50
Launched	66	52	55	44
Launched, approved CV drug verified in ChEMBL: strict rule	35	32	35	26
Launched, drug ever approved by FDA or EMA for a CV indication	33	30	33	24
Launched, approved CV drug verified in ChEMBL: permissive rule	58	46	49	40

Primary set: every indication that Minikel et al. tag as cardiovascular, plus indications matching eight lipid/atherosclerosis tokens. Sensitivity 1: the originally analysed set (MeSH branch C14 plus the lipid tokens). Sensitivity 2: sensitivity 1 plus cardiovascular-tagged terms filed under signs and symptoms. Sensitivity 3: MeSH branch C14 only.

Phase is the highest Minikel phase over the target's indications in the same set. Percentages are of all targets in the set.

Instrumented: at least one eQTLGen whole-blood cis-eQTL at $P<5\times10^{-8}$ within the gene body ±100 kb. No eQTL data: the gene is not among the 19,250 genes eQTLGen tested; its sub-types are autosomal genes not tested, and X-linked and mitochondrial genes, which eQTLGen did not analyse (no target is Y-linked). The autosomal-only lower bound is autosomal no-eQTL-data targets over targets not on X, Y or MT. No qualifying instrument: tested, but no variant meets the criterion. The upper bound is no eQTL data plus no qualifying instrument and is reported only as a bound.

The evidence rows on which the two ChEMBL rules disagree were reviewed and the strict rule was retained. Ever approved by FDA or EMA: the strict rule, and at least one of the target's strict-rule drugs has an approval record in Drugs@FDA or the EMA medicines table; this is the definition of 'approved' used in the text. Two of the 33 rest on an indication that is cardiovascular only in a broad sense: NR3C1 (dexamethasone, prednisolone and prednisone; ChEMBL approved indication myocarditis) and MS4A1 (rituximab; systemic vasculitis). Both are instrumented, so they enter the denominator only. Strict: an approved drug linked directly to the gene (single protein, specific complex or mRNA) has an approved indication that sits only in cardiovascular MeSH branches. Permissive: any ChEMBL link, any approved indication under MeSH C14 or a lipid term.

Table 2. Candidate instrument availability for the 13 launched targets with an approved cardiovascular drug in ChEMBL and no eQTLGen whole-blood instrument

Target	Approved cardiovascular drug(s) in ChEMBL	Site of drug action	Regulatory status (FDA / EMA)	eQTLGen status	Candidate-instrument tier	GTEx tissues with instrument, n	GTEx tissues with instrument (lead P)	GTEx whole-blood instrument	UKB-PPP cis-pQTL	deCODE cis-pQTL	Fenland cis-pQTL
AGTR1	azilsartan medoxomil; candesartan cilexetil; eprosartan; irbesartan; losartan; olmesartan medoxomil; telmisartan; valsartan	Angiotensin II type 1 receptor: vasoconstriction and aldosterone secretion by the adrenal cortex (FDA label, losartan)	Marketed: 7 of 8 drugs; discontinued: eprosartan (FDA discontinued)	No eQTL data: autosomal, not tested	CV-relevant or plasma-protein candidate	6	Cultured fibroblasts (5.8×10^{-41}); Nerve Tibial (1.4×10^{-14}); Artery Tibial* (3×10^{-14}); Heart Atrial Appendage* (7.7×10^{-14}); Colon Sigmoid (2.7×10^{-13}); Adipose Subcutaneous (6.9×10^{-11})	No	Not assayed	Not assayed	Not assayed
ANGPTL3	evinacumab	ANGPTL3, expressed primarily in the liver; the antibody binds the circulating protein (FDA label, evinacumab)	Marketed: 1 of 1 drugs	No eQTL data: autosomal, not tested	CV-relevant or plasma-protein candidate	0	None	No	rs775677524, $P=1\times10^{-500}$	rs773045547, $P=6.03\times10^{-269}$	rs72649573, $P=2.6\times10^{-23}$
APOB‡	mipomersen	Cells that produce apolipoprotein B (antisense blocks its production); organ not stated (EMA summary, Kynamro)	No currently marketed drug found; discontinued: mipomersen (FDA discontinued; EMA refused)	No eQTL data: autosomal, not tested	CV-relevant or plasma-protein candidate	9	Skin Sun Exposed Lower leg (4.6×10^{-58}); Skin Not Sun Exposed Suprapubic (2.3×10^{-35}); Heart Left Ventricle* (1.1×10^{-30}); Adipose Subcutaneous (4.9×10^{-29}); Nerve Tibial (8.4×10^{-16}); Heart Atrial Appendage* (1.1×10^{-11}); Artery Tibial* (8×10^{-11}); Esophagus Gastroesophageal Junction (6.9×10^{-9}); Esophagus Muscularis (1.1×10^{-8})	No	Assayed, none listed	rs563290, $P=2\times10^{-13}$	rs541041, $P=1.05\times10^{-13}$
APOC3	volanesorsen	Production of apolipoprotein C-III (antisense); organ not stated (EMA summary, Waylivra)	Marketed: 1 of 1 drugs	No eQTL data: autosomal, not tested	CV-relevant or plasma-protein candidate	0	None	No	Not assayed	rs964184, $P=2.95\times10^{-63}$	rs964184, $P=8.39\times10^{-27}$
AVPR2 (X)	conivaptan; tolvaptan	Vasopressin V2 receptor; effect is renal water excretion (FDA label, tolvaptan)	Marketed: 2 of 2 drugs	No eQTL data: X-linked, not analysed	Locked: never assayed	0	None	No	Not assayed	Not assayed	Not assayed
F2	argatroban; bivalirudin; dabigatran etexilate; desirudin; lepirudin; ximelagatran	Circulating thrombin, at its active site (FDA label, argatroban)	Marketed: 3 of 6 drugs; discontinued: desirudin (FDA discontinued); not confirmed: lepirudin (no FDA record; EMA: withdrawn), ximelagatran (no FDA record; EMA: application withdrawn)	No eQTL data: autosomal, not tested	CV-relevant or plasma-protein candidate	3	Artery Tibial* (3.3×10^{-15}); Brain Cerebellum (2×10^{-12}); Testis (1.1×10^{-9})	No	rs1799963, $P=6.3\times10^{-127}$	rs3136516, $P=4.68\times10^{-41}$	rs1799963, $P=2.4\times10^{-33}$
PCSK9	alirocumab;	Circulating PCSK9; LDL	Marketed: 3 of 3	No eQTL data:	CV-relevant or	6	Nerve Tibial (2.5×10^{-64}); Adipose	Yes	rs1159114	rs1159114	rs1159114

Target	Approved cardiovascular drug(s) in ChEMBL	Site of drug action	Regulatory status (FDA / EMA)	eQTLGen status	Candidate-instrument tier	GTEx tissues with instrument, n	GTEx tissues with instrument (lead P)	GTEx whole-blood instrument	UKB-PPP cis-pQTL	deCODE cis-pQTL	Fenland cis-pQTL
	evolocumab; inclisiran	receptors on hepatocytes (FDA label, alirocumab)	drugs	autosomal, not tested	plasma-protein candidate		Visceral Omentum (2.3×10^{-28}); Lung (4.4×10^{-20}); Testis (1.8×10^{-18}); Brain Cerebellar Hemisphere (9.7×10^{-11}); Brain Cerebellum (2.1×10^{-10})		7, $P=1\times10^{-502}$	7, $P=3.98\times10^{-117}$	7, $P=2.41\times10^{-64}$
PLG§	anistreplase; streptokinase	Not available from the sources checked	Current marketing not confirmed in either source; not confirmed: anistreplase (no FDA or EMA central record), streptokinase (no FDA or EMA central record)	No eQTL data: autosomal, not tested	CV-relevant or plasma-protein candidate	1	Testis (5.2×10^{-15})	No	rs73015965, $P=1\times10^{-550}$	rs143079629, $P=1.15\times10^{-159}$	rs537579467, $P=2.03\times10^{-178}$
REN	aliskiren	Renin, secreted by the kidney, acting on circulating angiotensinogen (FDA label, aliskiren)	Marketed: 1 of 1 drugs	No eQTL data: autosomal, not tested	CV-relevant or plasma-protein candidate	5	Skin Sun Exposed Lower leg (4.3×10^{-24}); Ovary (1.5×10^{-12}); Skin Not Sun Exposed Suprapubic (1.4×10^{-11}); Testis (5.9×10^{-11}); Brain Cortex (1.6×10^{-8})	No	rs193280350, $P=9.18\times10^{-128}$	rs193280350, $P=1.12\times10^{-103}$	rs141995914, $P=3.62\times10^{-34}$
SLC5A1	sotagliflozin	SGLT1 in the intestine (and SGLT2 in the kidney); label states the mechanism of cardiovascular benefit is not established (FDA label, sotagliflozin)	Marketed: 1 of 1 drugs	No eQTL data: autosomal, not tested	Other-tissue candidate only	3	Thyroid (8.5×10^{-18}); Spleen (4.7×10^{-14}); Nerve Tibial (1.8×10^{-9})	No	Not assayed	Not assayed	Not assayed
IL1B†	rilonacept	Soluble IL-1α and IL-1β, bound by a decoy receptor (FDA label, rilonacept)	Marketed: 1 of 1 drugs	No qualifying instrument: significant cis-eQTL only outside ±100 kb	Other-tissue candidate only	2	Testis (6.8×10^{-26}); Skin Sun Exposed Lower leg (4.7×10^{-11})	No	Assayed, none listed	Assayed, none listed	Assayed, none listed
ITGA2B	abciximab; eptifibatide; tirofiban	Platelet glycoprotein IIb/IIIa (FDA label, eptifibatide)	Marketed: 2 of 3 drugs; discontinued: abciximab (FDA discontinued)	No qualifying instrument: cis-eQTL at FDR<0.05, none at $P<5\times10^{-8}$	Other-tissue candidate only	1	Thyroid (7.6×10^{-17})	No	Not assayed	Not assayed	Assayed, none listed
NPC1L1	ezetimibe	Brush border of the small intestine (FDA label, ezetimibe)	Marketed: 1 of 1 drugs	No qualifying instrument: no cis-eQTL at FDR<0.05	Other-tissue candidate only	2	Adipose Subcutaneous (1.7×10^{-18}); Testis (9.2×10^{-10})	No	Not assayed	Not assayed	Not assayed

Targets are launched in the primary set and pass the strict drug-verification rule; drug links and approval records were verified (Table S18). Angiotensin II, an AGTR1 agonist approved for shock, is linked to AGTR1 in ChEMBL and is not listed. Instrument:

a variant at $P<5\times10^{-8}$ within the gene body ±100 kb in the resource's own build. GTEx v8 tissues other than whole blood are listed in order of lead p-value (in brackets); * marks the eight pre-specified cardiovascular-relevant tissues. pQTL columns give the lead qualifying cis variant in the published table of each study. (X) X-linked gene. Tiers describe candidate instrument availability, not validated instruments. CV-relevant or plasma-protein candidate: a qualifying variant in a cardiovascular-relevant tissue or a plasma cis-pQTL. Other-tissue candidate only: a qualifying variant only in other tissues. Locked: none in any checked resource; this is relative to the resources checked and is not a statement that no instrument exists. Site of drug action is taken from the mechanism-of-action text of an FDA label or an EMA public assessment summary; it was not available for PLG. Regulatory status: FDA marketing status from Drugs@FDA (openFDA) and status in the EMA medicines table, both accessed October 2026; drug-by-drug records and sources are in Supplementary Table S18. 'Marketed' means at least one product with FDA status Prescription, or EMA status Authorised; nationally authorised European products are not in the EMA table. † IL1B: the approved drug, rilonacept, binds both IL-1α and IL-1β (FDA label), and its approved cardiovascular indication in ChEMBL is pericarditis, whereas the source data label the target launched for atherosclerosis. IL1B is kept in this table and is not used as a named example elsewhere. ‡ APOB: the only linked drug, mipomersen (Kynamro), is listed as discontinued by the FDA and was refused marketing authorisation by the EMA in 2013. It was therefore approved at one time (by the FDA) and is no longer marketed; APOB remains a named example on that basis. § PLG: neither linked drug (anistreplase, streptokinase) has a record in Drugs@FDA or the EMA medicines table, so approval by either agency could not be confirmed from these sources. PLG is kept in this table and is not used as a named example elsewhere. P values below 1×10^{-300} are derived from the −log10 P reported in the source table, rounded to the nearest power of 10. 'Approved' in this study means ever approved by the FDA or the EMA for a cardiovascular indication; the cardiovascular indication is taken from ChEMBL and the approval record from the two agencies' databases.

Table 3. Recovery map of candidate instrument availability for the 280 cardiovascular targets with no eQTLGen whole-blood instrument

	Targets	Candidate: CV-relevant tissue or plasma protein	Candidate: other tissue only	Locked, total	Locked: never assayed	Locked: assayed, no instrument
All uninstrumentable targets	280	151 (53.9%)	72 (25.7%)	57 (20.4%)	35 (12.5%)	22 (7.9%)
By eQTLGen status						
No eQTL data: autosomal, not tested	188	105 (55.9%)	46 (24.5%)	37 (19.7%)	25 (13.3%)	12 (6.4%)
No eQTL data: X-linked, not analysed	27	15 (55.6%)	8 (29.6%)	4 (14.8%)	3 (11.1%)	1 (3.7%)
No eQTL data: mitochondrial, not analysed	2	0 (0.0%)	0 (0.0%)	2 (100.0%)	2 (100.0%)	0 (0.0%)
No qualifying instrument	63	31 (49.2%)	18 (28.6%)	14 (22.2%)	5 (7.9%)	9 (14.3%)
By highest clinical phase						
Preclinical	126	65 (51.6%)	27 (21.4%)	34 (27.0%)	21 (16.7%)	13 (10.3%)
Phase I	36	18 (50.0%)	12 (33.3%)	6 (16.7%)	3 (8.3%)	3 (8.3%)
Phase II	66	35 (53.0%)	21 (31.8%)	10 (15.2%)	7 (10.6%)	3 (4.5%)
Phase III	25	16 (64.0%)	3 (12.0%)	6 (24.0%)	3 (12.0%)	3 (12.0%)
Launched	27	17 (63.0%)	9 (33.3%)	1 (3.7%)	1 (3.7%)	0 (0.0%)

	Targets	Candidate: CV-relevant tissue or plasma protein	Candidate: other tissue only	Locked, total	Locked: never assayed	Locked: assayed, no instrument
By indication area						
Lipids / atherosclerosis	71	41 (57.7%)	16 (22.5%)	14 (19.7%)	7 (9.9%)	7 (9.9%)
Ischaemic heart disease	65	41 (63.1%)	15 (23.1%)	9 (13.8%)	7 (10.8%)	2 (3.1%)
Heart failure / cardiomyopathy	70	25 (35.7%)	24 (34.3%)	21 (30.0%)	16 (22.9%)	5 (7.1%)
Arrhythmia	18	11 (61.1%)	4 (22.2%)	3 (16.7%)	3 (16.7%)	0 (0.0%)
Hypertension	31	15 (48.4%)	9 (29.0%)	7 (22.6%)	6 (19.4%)	1 (3.2%)
Pulmonary hypertension	38	17 (44.7%)	12 (31.6%)	9 (23.7%)	4 (10.5%)	5 (13.2%)
Thrombosis	22	17 (77.3%)	5 (22.7%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Cerebrovascular	72	43 (59.7%)	16 (22.2%)	13 (18.1%)	9 (12.5%)	4 (5.6%)
Other	94	60 (63.8%)	21 (22.3%)	13 (13.8%)	6 (6.4%)	7 (7.4%)
Unspecified CV	57	30 (52.6%)	19 (33.3%)	8 (14.0%)	4 (7.0%)	4 (7.0%)

Resources checked: GTEx v8 cis-eQTLs in 48 tissues other than whole blood, and the published cis-pQTL tables of UKB-PPP, deCODE and Fenland. A candidate instrument is a variant at $P<5\times10^{-8}$ within the gene body ±100 kb; its validity as an instrument was not tested. Candidate, CV-relevant tissue or plasma protein: a candidate in one of eight pre-specified cardiovascular-relevant tissues (heart left ventricle, atrial appendage, aorta, coronary artery, tibial artery, liver, kidney cortex, adrenal gland) or a plasma cis-pQTL. Candidate, other tissue only: a candidate only in other tissues. Locked: no candidate in any checked resource, split by whether any of the three proteomic studies measured the

protein. The tiers were defined after the first results were seen. A target is counted in every indication area in which it has a programme, so area rows do not sum to the total. Percentages are of the row. Never-assayed counts include microRNA and mitochondrial genes, which no plasma protein panel can measure (9 and 2 overall; 4 and 1 among heart failure and cardiomyopathy targets), so they exceed the protein-coding counts given in the text.

Figures

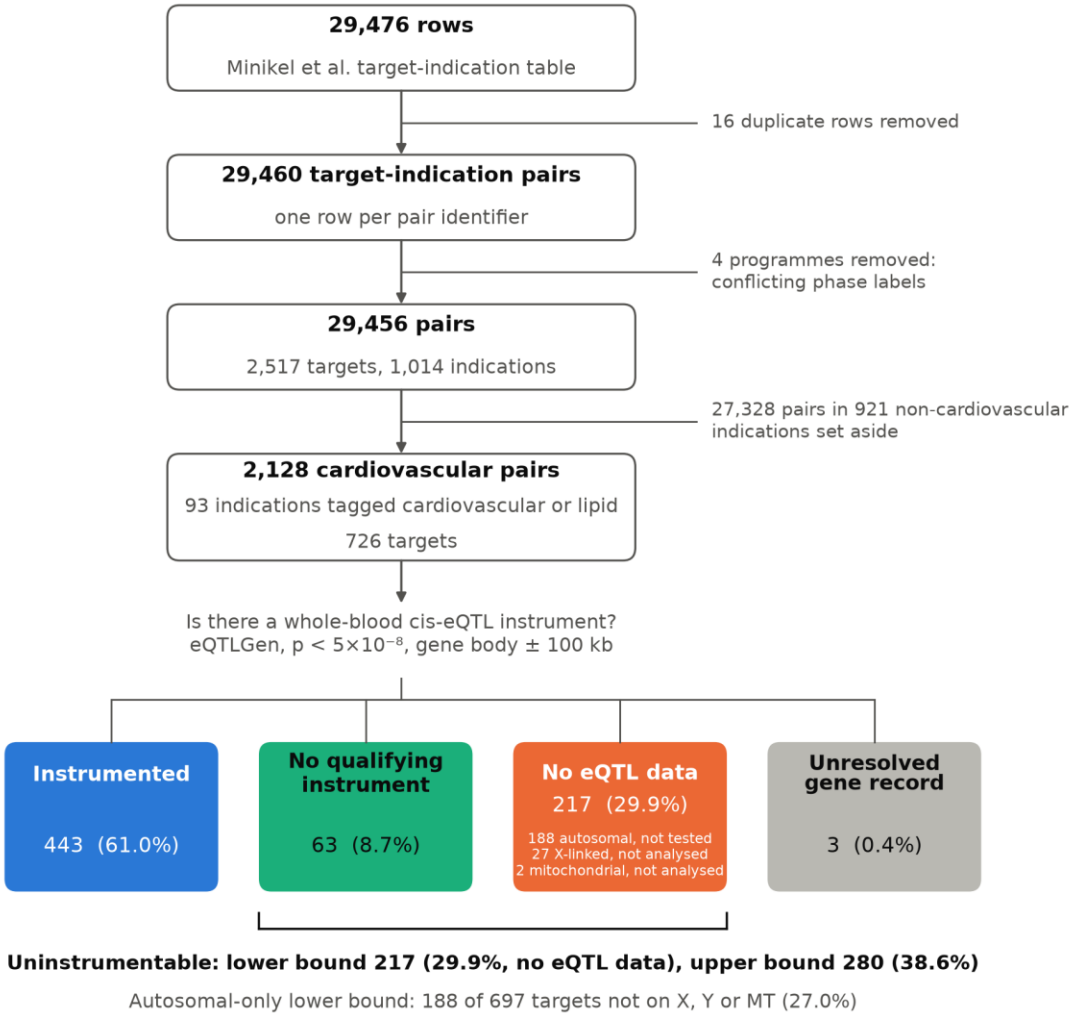


Figure 1

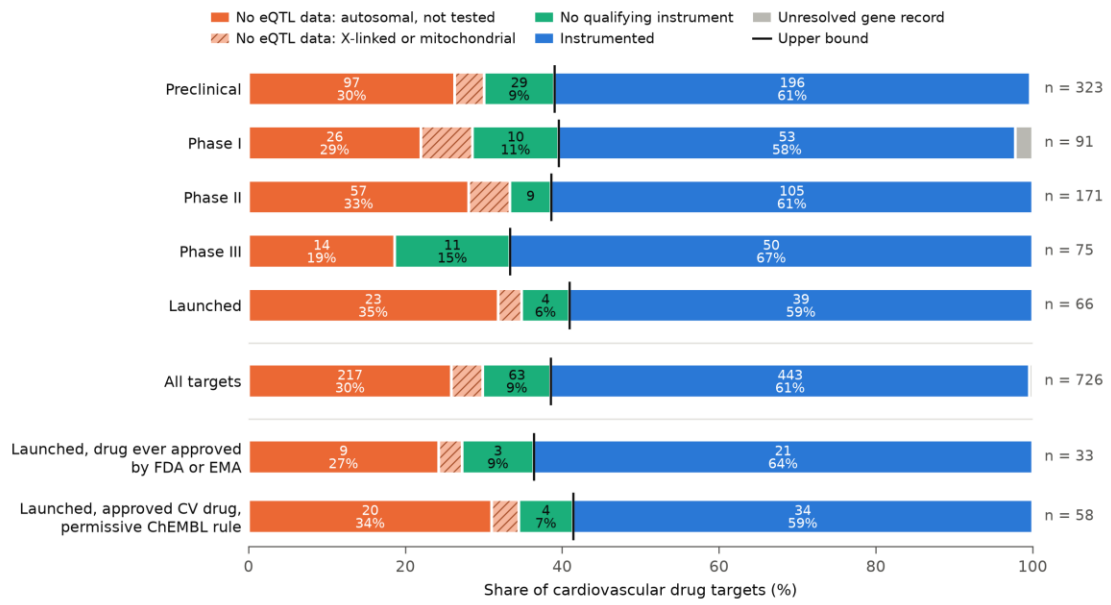


Figure 2

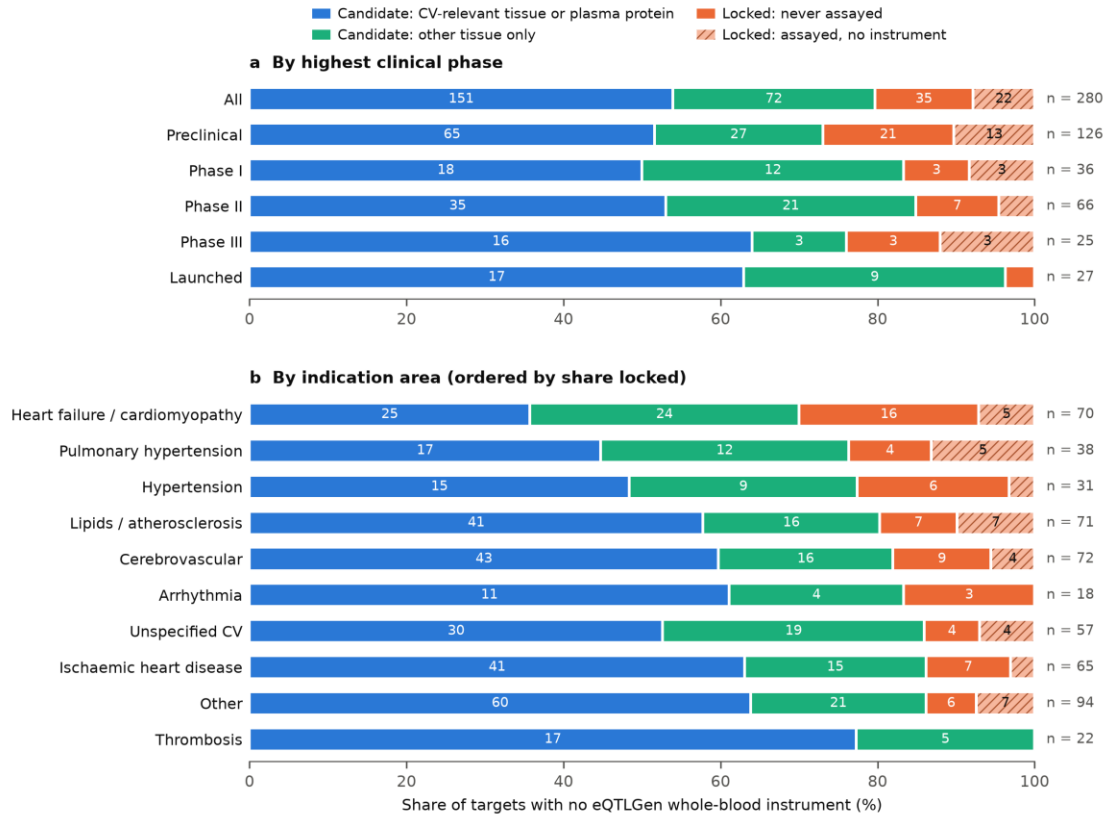


Figure 3